

Clinical Risk Assessment Report: Patient 55

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 33.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.51x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.8%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 2.18 [Top 4%] (+5.8%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.16 [Top 12%] (+2.7%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.69 [Avg (28th %ile)] (+2.5%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.21 [Bottom 16%] (+2.3%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.76 [Top 4%] (+1.8%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

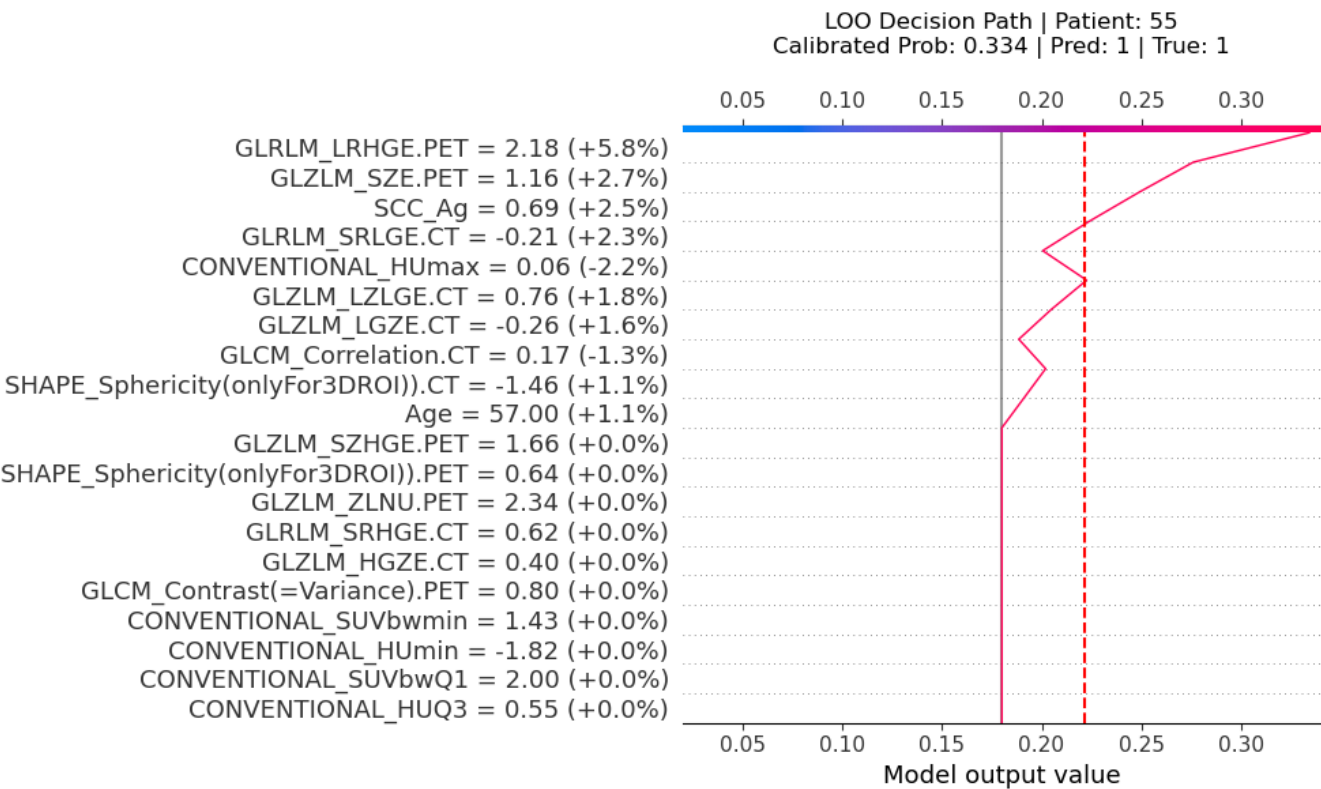
Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.26 [Avg (35th %ile)] (+1.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

3. Protective / Mitigating Factors (Reducing Risk)

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.06 [Avg (47th %ile)] (-2.2%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.17 [Avg (54th %ile)] (-1.3%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

